Abalone Age Prediction using Machine Learning Algorithms

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DAP391m AI-DS Project

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1. Abstract

Traditional abalone age determination is a destructive and labor-intensive process, requiring specimens to be harvested and sectioned for microscopic ring counting. This study proposes a non-invasive machine learning alternative using physical features from the UCI Abalone dataset. By evaluating OLS, Ridge, and SVR architectures within a robust preprocessing pipeline, we identify OLS as the most stable model for deployment. Achieving an RMSE of 2.1818, this approach offers a transparent and computationally efficient solution for sustainable aquaculture management without the need for invasive sampling.

2. Introduction

To predict the number of physical rings in abalones as a proxy for estimating biological age based on physical characteristics.Traditional age determination, which involves cutting the shell and counting rings under a microscope, is both labor-intensive and expensive. Implementing Machine Learning facilitates a rapid, non-invasive, and cost-effective alternative.

3. Related Work

In the study *"Prediction of the age of abalones based on machine learning algorithms"* by Muzi Li (2022) [1], the author achieved an impressive RMSE of 1.44 using the Random Forest (RF) algorithm. This ensemble method proved superior in capturing complex non-linear relationships and high-order interactions among the physical variables of abalones.

3.1 Data Description

Data Source: The dataset was retrieved from the UCI Machine Learning Repository, comprising 4,177 observations [2].

Features:

> Sex: Categorical (M: Male, F: Female, I: Infant). Length: Longest shell measurement (mm). Diameter: Measurement perpendicular to length (mm).

Height: Meat height within the shell (mm). Weights: Whole weight, Shucked weight, Viscera weight, and Shell weight (grams).

Target Variable: Rings (Integer value). The biological age is estimated as: Age = Rings + 1.5

Feature Engineering: Size_Area: Calculated as Length * Diameter to represent the surface area, enabling the model to better capture the area-weight correlation.

4. Main Contribution

4.1 Problem Formulation

The task is defined as a Multiple Regression problem. The goal is to determine a mapping function f(x) that minimizes the error between the predicted value $\hat{y}$ and the ground truth y (number of rings).

Target: $y \in Z^+$ (Rings).

Input: Matrix X containing physical features and encoded sex categories.

4.2 EDA Insights

Correlation Heatmap: Analysis indicates that Shell_weight (r=0.63) and Diameter (r=0.57) are the strongest predictors. However, extreme multicollinearity (r > 0.9) among weight variables necessitated the creation of the Size_Area feature.

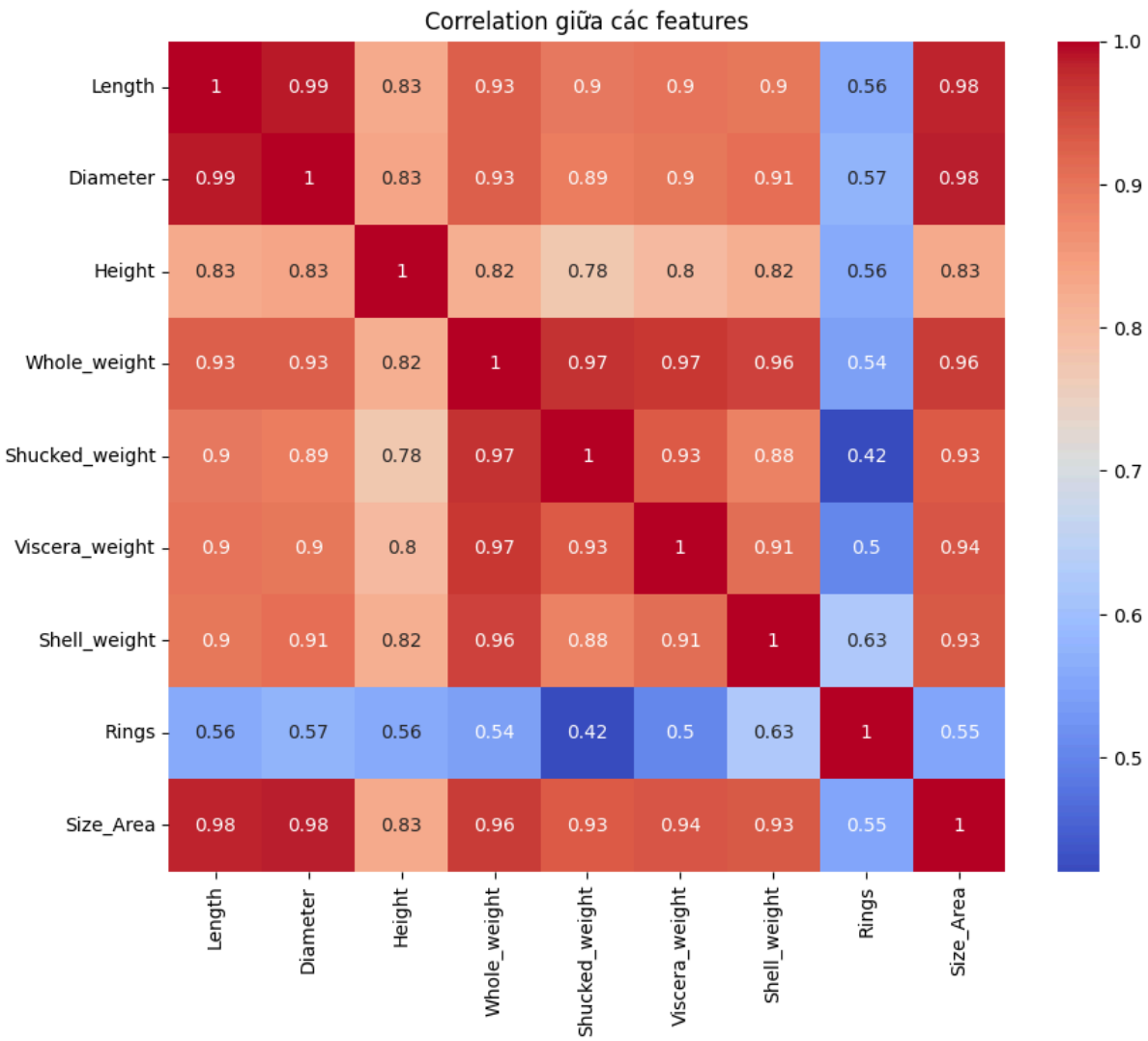

Figure 1. Correlation Heatmap

Scatter Plot (Size_Area vs. Rings): This derivative feature smooths the linear relationship with Rings, effectively separating overlapping data clusters.

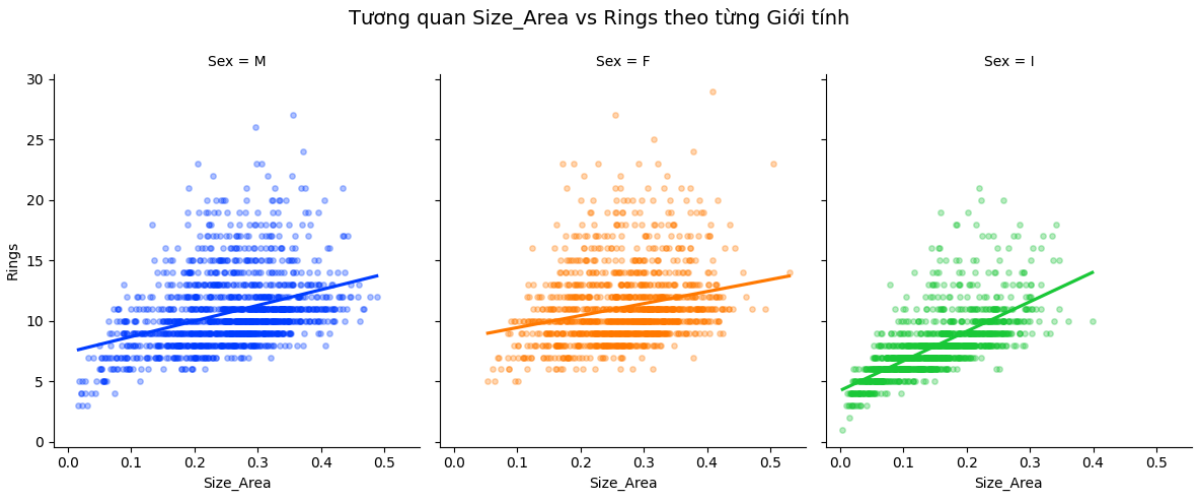

Figure 2. Sample Distribution across Ring Counts

Whole Weight vs. Rings by Sex: The Infant (I) group dominates the low-ring spectrum (under 10) with a stable linear growth trend. Conversely, mature groups (M and F) exhibit significantly higher variance once weight exceeds 1.5g, explaining the increased error rates in older specimens.

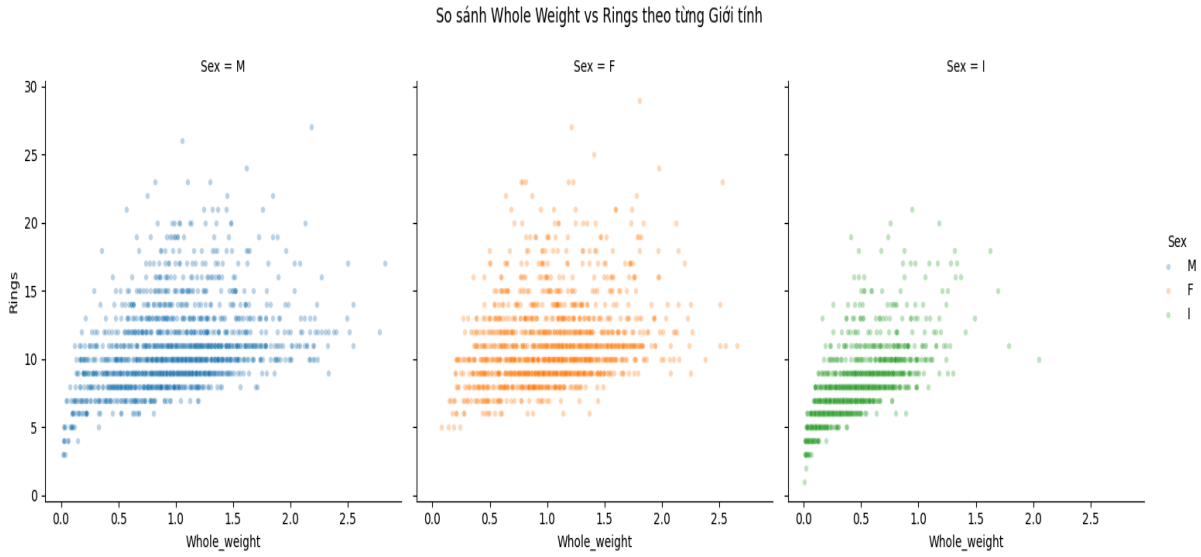

Figure 3. Whole Weight vs. Rings Correlation by Sex

Sex vs. Rings Distribution: The Infant group shows the lowest median (approx 7-8) and minimal variance, whereas mature groups display right-skewed distributions with heavy overlap between 8-15 rings.

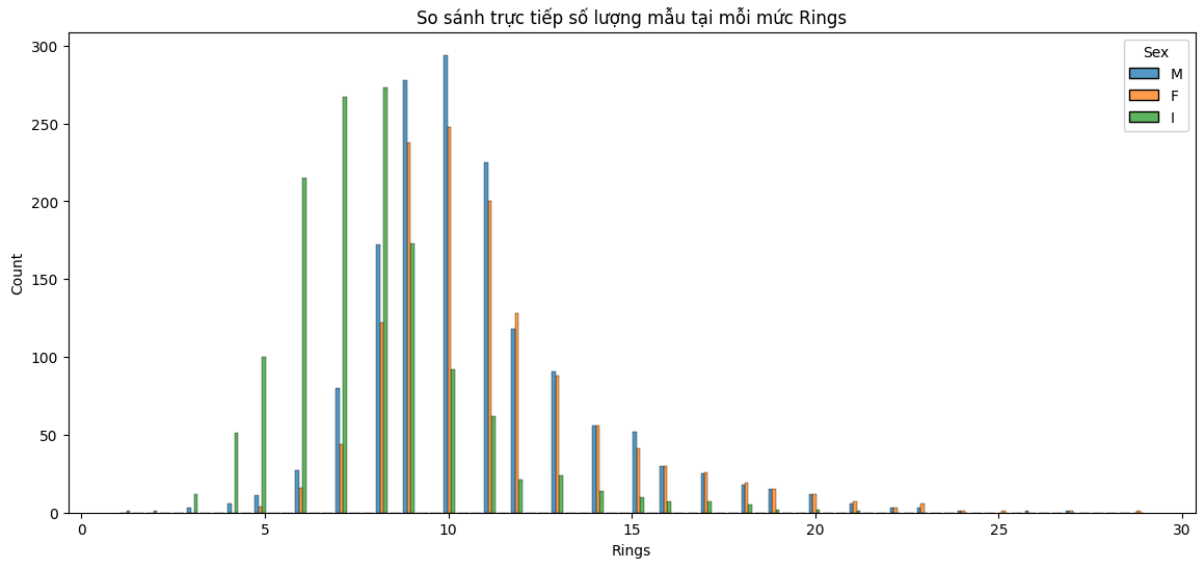

Figure 4. Sex vs. Rings Distribution (Boxplot)

4.3 Data Engineering

Outlier Checking: Boxplots revealed significant outliers in weight variables. We elected to retain these samples to preserve population integrity and test model robustness against noise.

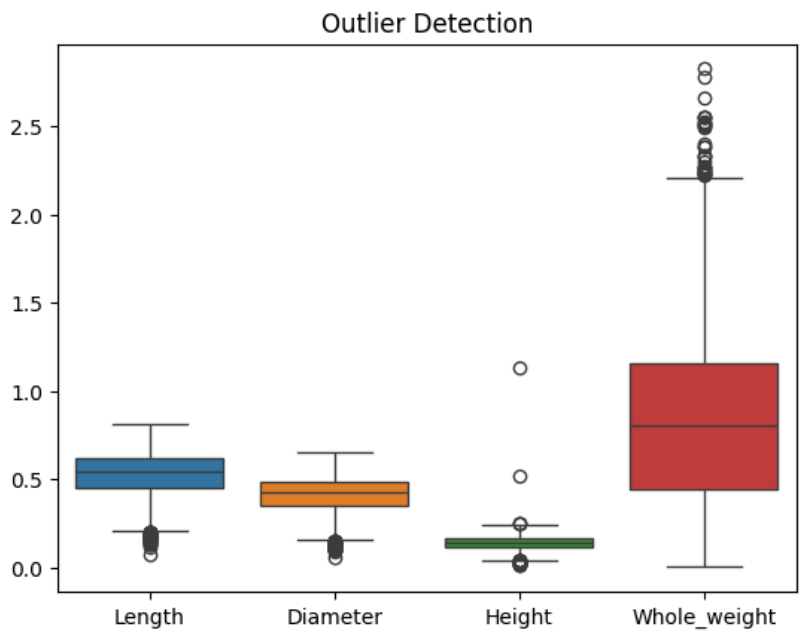

Figure 5. Strong parameter outlier check

Data Cleaning: Samples with Rings = 1 were excluded due to extreme rarity and lack of statistical representation, which could bias the loss function.Robust Scaling: Given the presence of outliers, RobustScaler was utilized instead of Standard Scaler:

$$x_{Scaled} = \frac{x_i - Q_2(x)}{Q_3(x) - Q_1(x)}$$

This method leverages the Median and Interquartile Range (IQR), ensuring the scaling is not skewed by extreme values.

4.4 Model Selection and Definition

We implemented a three-stage experimental roadmap with increasing complexity:

1. Ordinary Least Squares (OLS): Used as a reference point to evaluate the fundamental linear relationships. To minimize the sum of squared residuals (differences between the observed and predicted values):

$$\min_{\beta} \|y - X\beta\|^2$$

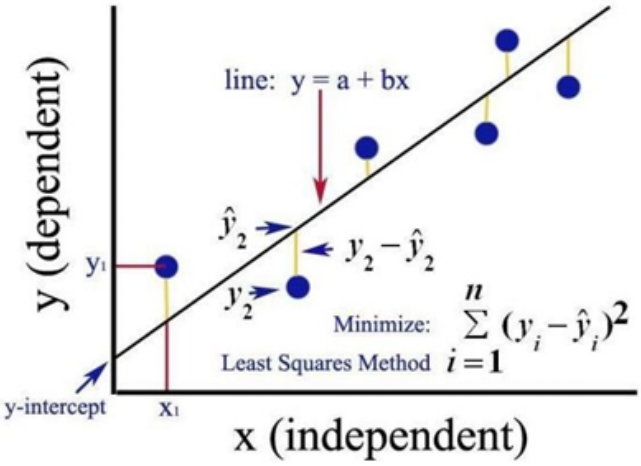

Figure 6. Linear regression schematic [6]

2. Ridge Regression:

Addressed multicollinearity highlighted in EDA by applying an penalty.[3]To mitigate multicollinearity by introducing an L2 regularization term, which penalizes large coefficients and reduces model variance:

$$\min_{\beta}(||y - X\beta||_2^2 + \alpha||\beta||_2^2)$$

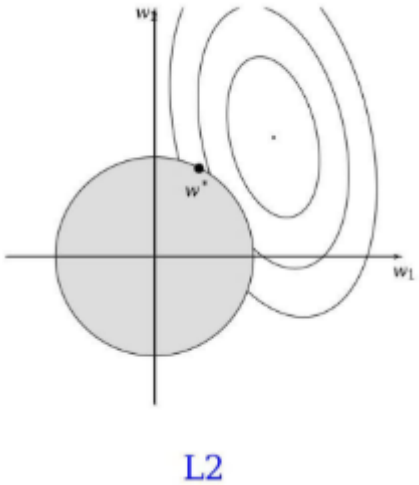

Figure 7. Geometric Interpretation of L2 Regularization in Ridge Regression.[7]

3. Support Vector Regression (SVR):

Utilized the RBF Kernel to map data into higher-dimensional space to capture non-linear growth patterns.Minimize the epsilon-insensitive tube:

$$\min \tfrac{1}{2}||\omega||^2 + C\sum_{i=1}^{n}(\xi_i + \xi_i^*)$$

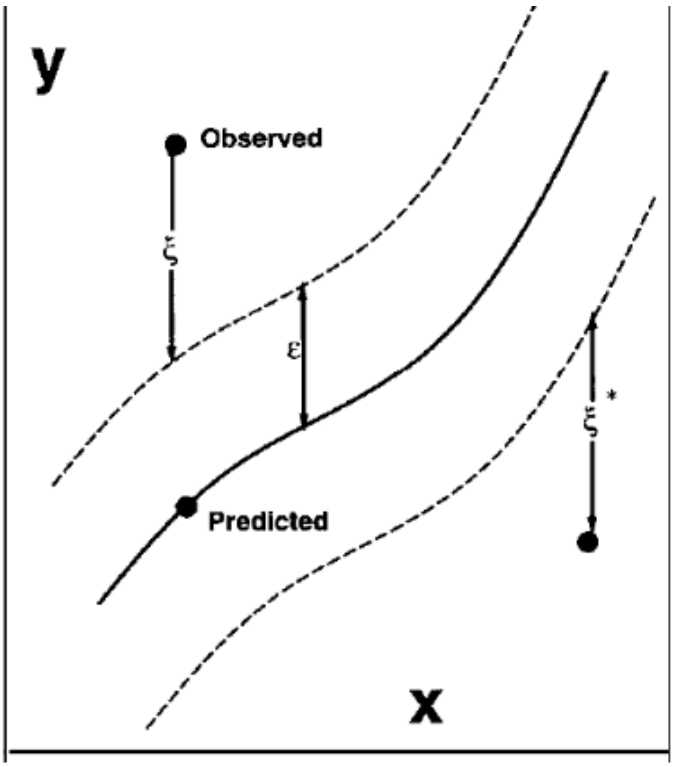

Figure 8. The parameter for the linear support vector regression.[8]

4.5 Training and Testing Strategy

Performance was validated using 5-Fold Cross-Validation [5] with these metrics. RMSE: Measures unit-level deviation [4]. R^2: Measures the proportion of variance explained by the model. MAE: Measures average absolute error magnitude.

5. Experiment Results

5.1 Tuning & Final Evaluation

Tuning Strategy: Optimized alpha for Ridge and $\{C, \epsilon\}$ for SVR via GridSearchCV. Findings: A notable phenomenon was observed: while SVR achieved the highest CV scores during training, OLS demonstrated superior stability and lower RMSE (2.1818) on the independent test set. Explanation**:** The inherent simplicity of OLS prevented the overfitting trap that affected SVR. The high degree of fundamental linearity in the abalone dataset makes OLS more robust against noise in unseen data.

```
--- FINAL COMPARISON ---
                    Model       MAE      RMSE        R2
0  OLS Linear Regression  1.566608  2.181804  0.582822
1           Ridge (Tuned)  1.566629  2.181845  0.582806
2             SVR (Tuned)  1.510797  2.195423  0.577598

Best Model: OLS Linear Regression
```

Figure 9. Finale Model Comparison (R2 - RMSE - MAE)

6. Discussion

6.1 Error Analysis

By Sex: The lowest error was recorded in the Infant group due to highly consistent early-stage growth patterns.

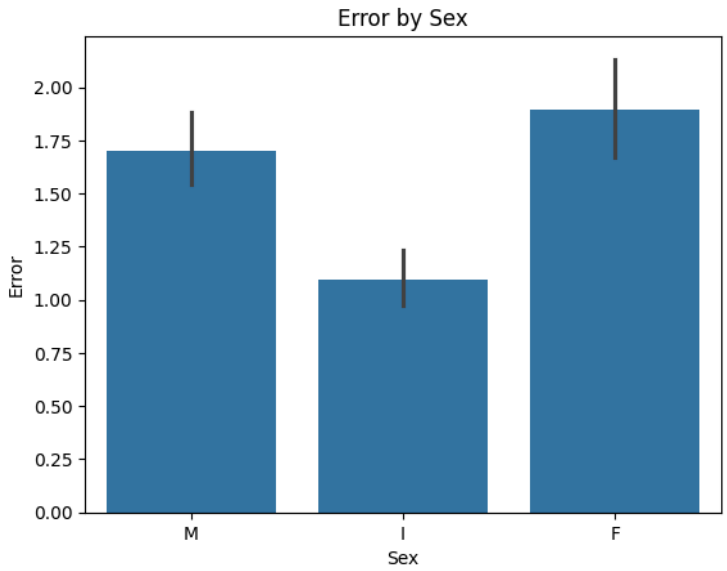

Figure 10. Error by Sex

By Ring Bins**:** The model is highly accurate in the 8-12 ring range. However, it tends to under-predict for older abalones (>15 rings) due to data sparsity in that segment.

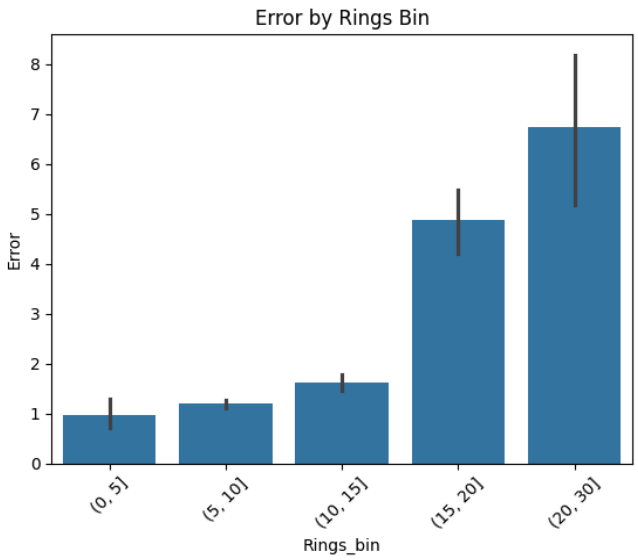

Figure 11. Error by Rings Bin

6.2 Visualization Evaluation

Predicted vs. Actual: Samples are densely clustered around the 45-degree identity line at lower ring counts. At higher counts, a spread out effect appears, reflecting the biological volatility associated with aging.

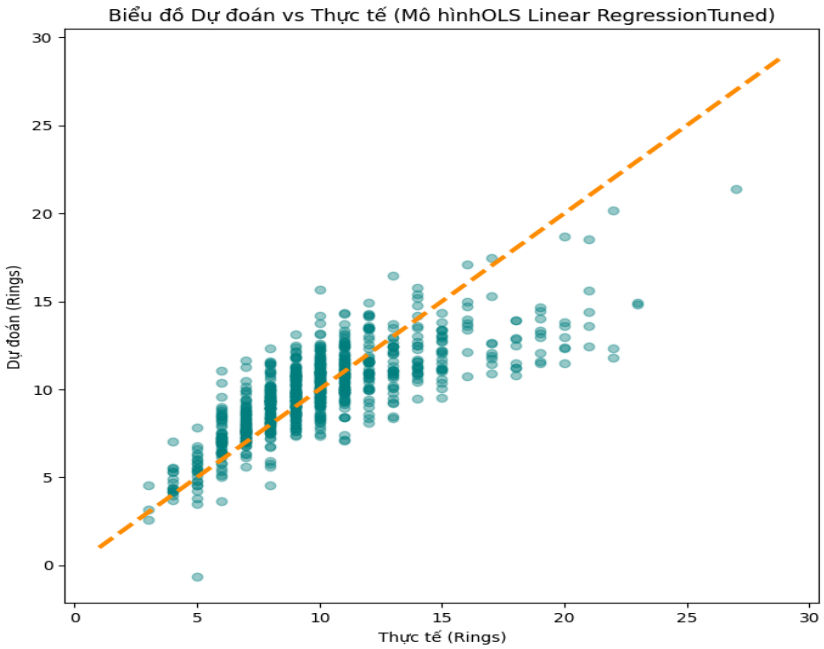

Figure 12. Predict and Actual Ring Scatter plot

The Residual Plot: displays a relatively random scatter around the zero-horizontal line at lower ring counts (under 12), suggesting a good fit for younger specimens. However, a noticeable funnel shape and rightward spread emerge for older abalones, indicating Heteroscedasticity (non-constant variance).

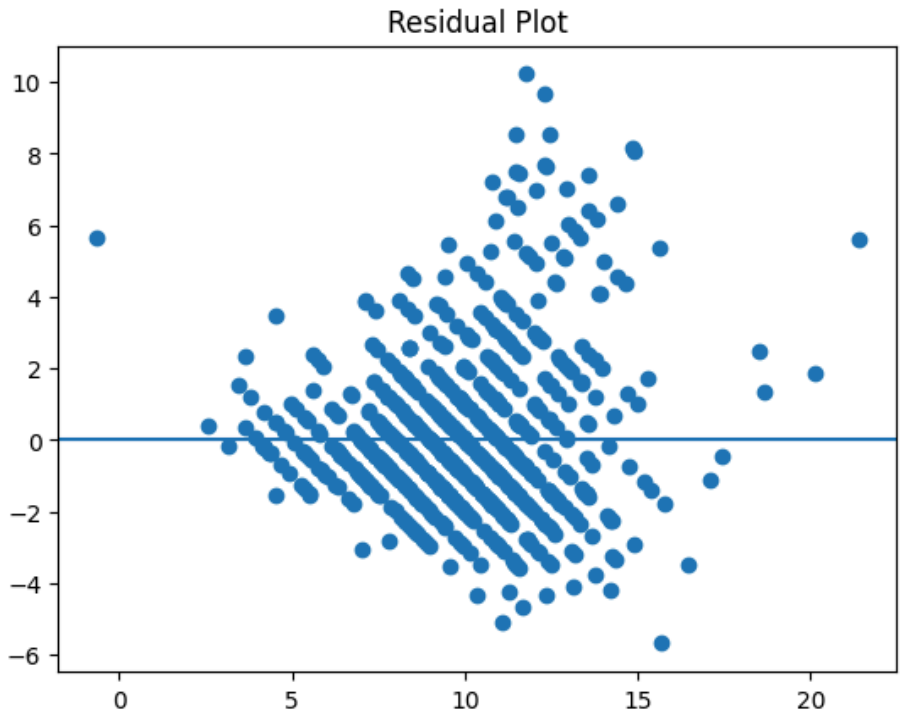

Figure 13. Residual Plot of the OLS Model

The Top Coefficients**:** bar chart reveals that Whole_weight and Diameter are the dominant positive predictors of Rings, while Shucked_weight has a strong negative correlation, potentially reflecting physiological energy trade-offs. The newly engineered Size_Area holds substantial predictive power, confirming that geometry-derived metrics are statistically viable for abalone age modeling.

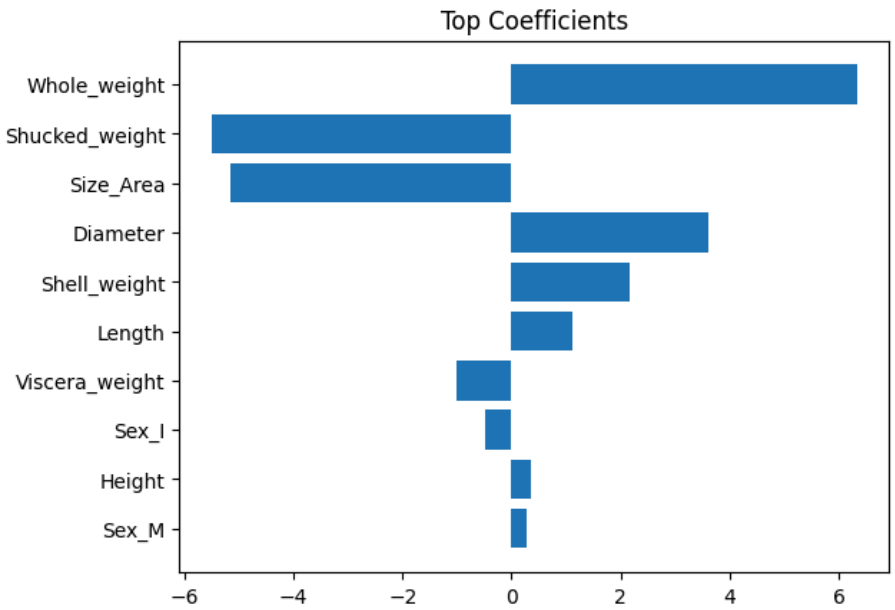

Figure 14. Top Coefficients for Age Prediction

6.3 Practical Impact

The OLS Pipeline was exported (joblib) for deployment. Choosing OLS ensures high-speed inference and instant feedback with an impressive accuracy within +- 2 rings, making it ideal for field-testing devices.

6.4 Model Performance and Trade-offs

When benchmarked against the RMSE of 1.44 achieved by Muzi Li (2024) using Random Forest, our OLS model yielded an RMSE of 2.1818.[1]. This performance variance can be elucidated through the following technical perspectives:

```
Accuracy (±1 ring): 0.444311377245509
Accuracy (±2 rings): 0.7437125748502994
Accuracy (±2 rings): 0.8754491017964072
```

Figure 15. Finale Accuracy of our project

Table 1. Abalone age prediction results.

Model	Discrepancy
Linear regression	1.80 years old
Decision tree	1.61 years old
Random Forest	1.44 years old
Support vector machine	1.64 years old

Figure 16. Finale discrepancy of Muzi Li project [1]

Model Complexity and Interpretability: While Random Forest is an ensemble-based algorithm capable of capturing highly intricate, non-linear data structures, we deliberately opted for Ordinary Least Squares (OLS). Our objective prioritized interpretability (White-box modeling) and computational efficiency, ensuring that the relationship between physical features and age remains transparent and mathematically explicit. Mitigation of Overfitting: Complex non-linear models like Random Forest are often susceptible to overfitting without rigorous regularization strategies. By selecting OLS, we provide a fundamental linear perspective on the feature space. This approach facilitates a lightweight deployment for demonstration systems, requiring significantly lower computational resources while maintaining robust performance. Research Autonomy and Pipeline Optimization: Rather than replicating existing high-complexity architectures, this study focuses on the autonomy of the preprocessing pipeline. By integrating RobustScaler, we demonstrate that even a foundational algorithm like OLS can achieve stable and reliable practical accuracy (within a margin of +-2 rings) when the data is properly engineered.

7. Conclusion

Recap: Developed a predictive system utilizing RobustScaler and OLS Regression. Limitations: The model currently struggles with the high-ring count group (>15 rings). Future Work: We propose testing Log-transformations on weight features to mitigate skewness and improve performance in high-variance segments.

8. References

[1] Li, M. (2024). *Prediction of the age of abalones based on machine learning algorithms*. Applied and Computational Engineering.

[2] Dua, D. & Graff, C. (2019). *UCI Machine Learning Repository*.

[3] Hoerl, A. E., & Kennard, R. W. (1970). *Ridge Regression*.

[4] Hastie, T., et al. (2009). *The Elements of Statistical Learning*.

[5] Kohavi, R. (1995). *A Study of Cross-Validation*.

[6] Linear Regression Schematic (Descending into ML): Linear Regression

[7] Hastie, T., Tibshirani, R., & Friedman, J. (2009). The Elements of Statistical Learning: Data Mining, Inference, and Prediction. Springer Science & Business Media.

[8] Drucker H, et al. 1997. Linear Support Vector Regression Machines.Advances in neural information processing systems, 155-161